

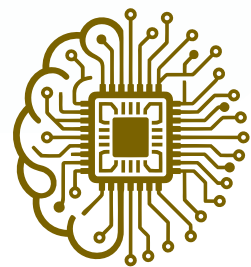


Capstone Projects

“From an idea in a notebook to a
solution in the real world.”



PRESENTED BY: QSTEMORAA CLUB/ CAPSTONE TEAM





What is a Capstone project?

- A project that discusses the main grand challenges that face the **World** generally and **Egypt** especially, and requires a solution with specific design requirements.

How many grades does the project have?

- The project weighs **%60** of the grades in total, also affecting the rest of the subjects with the same percentage of **%60**.

How to work on the project?

- Working on the project needs a team of **4-5 members** in Grade 10.
- Each member helps **equally** in researching, constructing, testing and presenting the project.
- The projects gets distributed to a number of categories: **Poster**, **Portfolio**, **Journal**, **Logbook**, and **Exhibition**.

How often do we work on a capstone project?

- There are 5 capstone projects with different challenges and requirements along the three years of high school in STEM, one each term of both **grade 10 and 11**, and one project for **grade 12**.

What to do when facing problems in the project?

- If it is a problem between **team members**; try to discuss it with each other and if there's no result yet, you can head to the capstone leader of your grade. If it is a **scientific problem**, you can ask your project monitors. If there's no solution yet, you can also head to the cap leader.



As we distributed parts of the project before, we can now discuss some details of each part: -

First: Portfolio: -

- a file or an **article** that collects all the information about your project: starting with members' information, going through the grand challenges and selection of solution, ending with the citation you searched for your project.
- The portfolio weighs **10%** of the project's grades, which means **6%** of the total grades.
- It's divided into **4 chapters**, each one resembling and discussing a stage of the project.
- Some parts of it are related to some parts of the poster. They are the same, and can be taken from one to another. (from the poster to the portfolio.)
- Each and every task of the portfolio must be distributed equally into the team members, and each team member must work with the same effort as the others.
- Portfolio is graded in the school, mainly depending on the local schoolteachers and cap leaders.



Second: Poster: -

- One large page with the sizes of 80 * 120cm presented alongside with the prototype in the exhibition day.
- Its content must be distributed and explained by each team member equally.
- The poster weighs about 40% of the total capstone grade, which means 24% of the total grade in grade 10 & 11, and 8% of the total grade in grade 12.
- It's divided into 11 parts: Abstract, Introduction, Materials & Methods, Test plan, Results, Analysis, Conclusion, Recommendations, Citation, Acknowledgement, and for further information.
- As mentioned before, some tasks are shared between the poster and the portfolio, but it's advised to take the task from the poster to the portfolio, not the other way around.
- Poster is graded by evaluators, and they have a copy of it to study before the exhibition day, so they can detect if there's any edits in the poster past the submission day.
- To avoid being caught editing the poster, you can inform them before starting to present the project, or start with the poster early to not need to edit it later.



Third: Journal: -

- A test of a certain number of questions that runs every fortnight.
- It has 3 questions in Grade 10, and 4 questions in both Grade 11 & 12. All questions must be solved in 1 hour at most.
- The journal weighs 30% of the capstone grades, and 18% of the total grade.
- The minimum number of words is 150, and the maximum is 250.
- Any journal in the three grades has three types of questions: Personal Reflection, Engineering Design Process, and Capstone Connection. But in grade 11 & 12, there are two capstone connection questions.
- Personal Reflection: A question that asks you to discuss your personal experience in the school, how you study and how do you collaborate with your time.
- Engineering Design Process: A question that asks you about your team's stage in the capstone at the moment, as for how you do your tasks, how you balance between your studies and the project, and how was your test plan and its results.
- Capstone Connection: A question that asks you to connect between the project's idea or test plan and a certain learning outcome of a certain subject that changes with every journal.



Third: Journal: -

- Journals are graded using four colors:
- **Blue:** When you answered the journal question in the best way possible, fulfilling all the questions requirements with adding examples and life experiences. **Blue = 4 GPA grade.** Ex/
- In (CH.1.09) you studied physical properties of different types of elements, Explain three properties for the material used in constructing your prototype to achieve the design requirements and two properties for a material you rejected as it doesn't help to achieve the design requirements. Learning Transfer
- It's actually surprising how such a concept actually helps in constructing the prototype, so we made sure that almost no energy is wasted because of the wrong choice of materials and their properties. Three properties adopted: - First: plasticity. This one is present in the filament used for printing the prototype, it can be pulled and shaped however we want without breaking down, which'll serve not collapsing while we are testing the prototype. Second: Strength. After printing the turbine, it looked soft (driving us for the third property) and delicate, but when applying vertical and horizontal pressure on it, we discovered that the turbine is actually stiff, holding up to 20 kilograms without breaking down, which means it can bear the ultimate-



Third: Journal: -

- high speeds of the winds caused by metro train the turbine will be applied in. Third: Texture. as mentioned before, the filament printed is pretty soft, which means there won't be energy wasted in friction of the rough surface, neither would it be harder to deal with when applying to the other components, to not scratch them. Two properties rejected: First: Elasticity. Elasticity means the material comes to its original shape after deforming, which we absolutely don't need for using filament, because it won't be shaped as the hybrid turbine we need. Second: Bounce. this concept of a property means how the material bounce on ground when thrown of a given height. We won't be throwing our turbine anyway, so we won't have advantage of using it.



Third: Journal: -

- **Green:** the grade you get when fulfilling the question's requirements with no additions or subtractions.
- **Green = 3.75 GPA - 2.75 GPA Ex/**

During the research step of the Engineering Design Process, explain how will studying existing methods, technologies, or real-world examples help you generate ideas, and improve your energy system design.

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Before making an idea and constructing a building for solving some grand challenges the world faces, especially the region the project will be applied in, it is a must to search up prior solutions, exploring its main idea, advantages, and disadvantages. This will help avoiding any ridiculous errors that'll end up destroying all the project.

For example, our big idea is to produce sustainable energy without causing much pollution, using recycled materials, and using a renewable energy, so I searched up a prior solution for the grand challenge I took, reducing air, water, and soil pollution.

I got inspired by the air pollution prevention action plan of China 2013, because it had clear goals, 10 majors to measure the success of their plan,



Third: Journal: -

and they mainly focused on awakening the population to the harsh consequences of air pollution, but it had the cons of not fully achieving the goal of getting the air pollution in their country less than the WHO limits.

Generally, the best way to arrange an idea is to search about the grand challenges, looking for the pre-existing solutions, studying the gaps they fell in when applying their idea as to not fall into them. This will make up a million dollar worth idea without going through much steps leading into nothing.

Yellow: The grade you get when not answering enough requirements or when answering with good grades and vocabulary but not matching the question.

Yellow = 2.75 GPA - 1.75 GPA

Red: The grade to get when detecting cheating, using AI, or not answering the question at all. When someone gets the grade and reports not cheating, there's a make up for the whole journal to get a better grade.

Red = 1.75 GPA - 0 GPA



Fourth: Log Book: -

- A PDF that every student makes at the ends of the capstone project for each semester to verify his/ her work along the semester.
- It's divided into 13/12 weeks, and each week has its own tasks that are written down in the calender.
- Each student must write down the tasks he finished for every week in the poster, portfolio, and prototype.
- The log book for eaach student must be agreed on for every team. In other words, one student must not write down a task he didn't finish or take it from his teammate.
- The logbook is signed down by the capstone leader for each graade, and the capstone General for the whole school.
- It must be printed and revised thoroughly in case the evaluators asked what task did you finish in a certain week.



Fifth: Exhibition day: -

- A day that comes at the ends of the semester in which we exhibit and present our project idea and the mechanisms of it.
- There are three rounds where three different evaluators grade and test your project with some questions, the number of rounds differs from one grade to another.
- The evaluators ask you about some parts of the poster, the mechanisms of your project, what sites did you use in searching for the idea.
- You get graded according to your presentations skills as an individual, collaboration skills as a team, and how convincing your project and test plan is.
- There are one evaluator that sits behind testing your collaboration skills as a team when there's no other evaluator that gives you a round of questions.